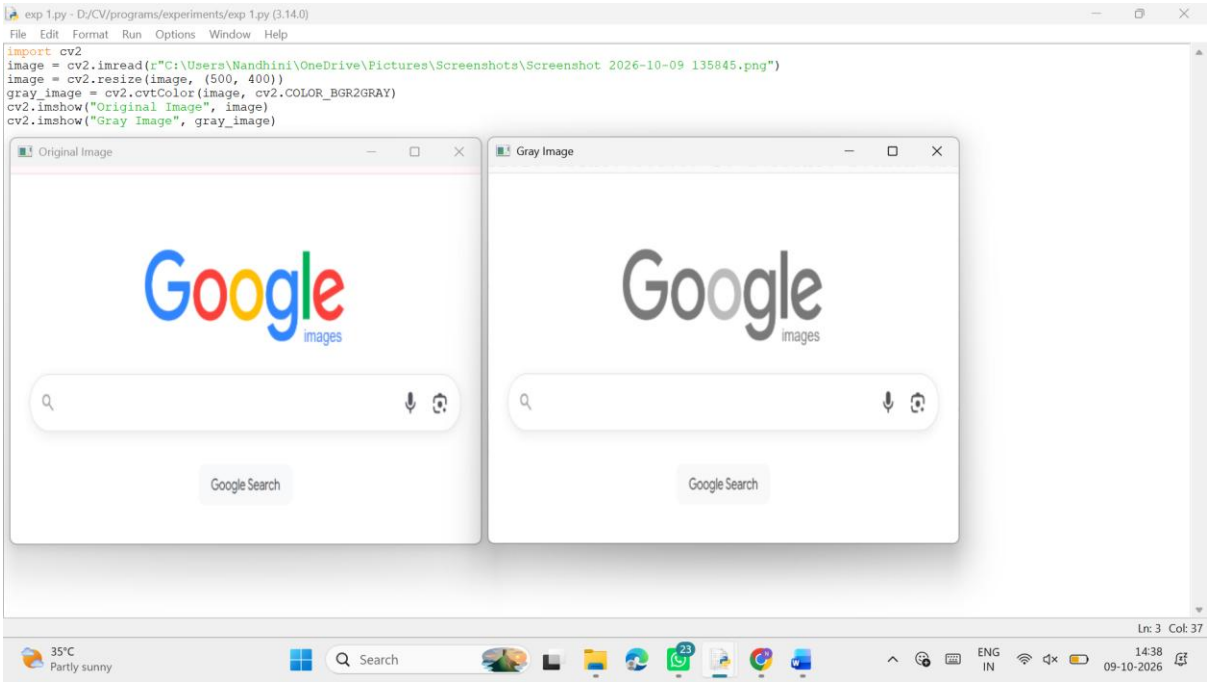
CV LAB EXPERIMENTS

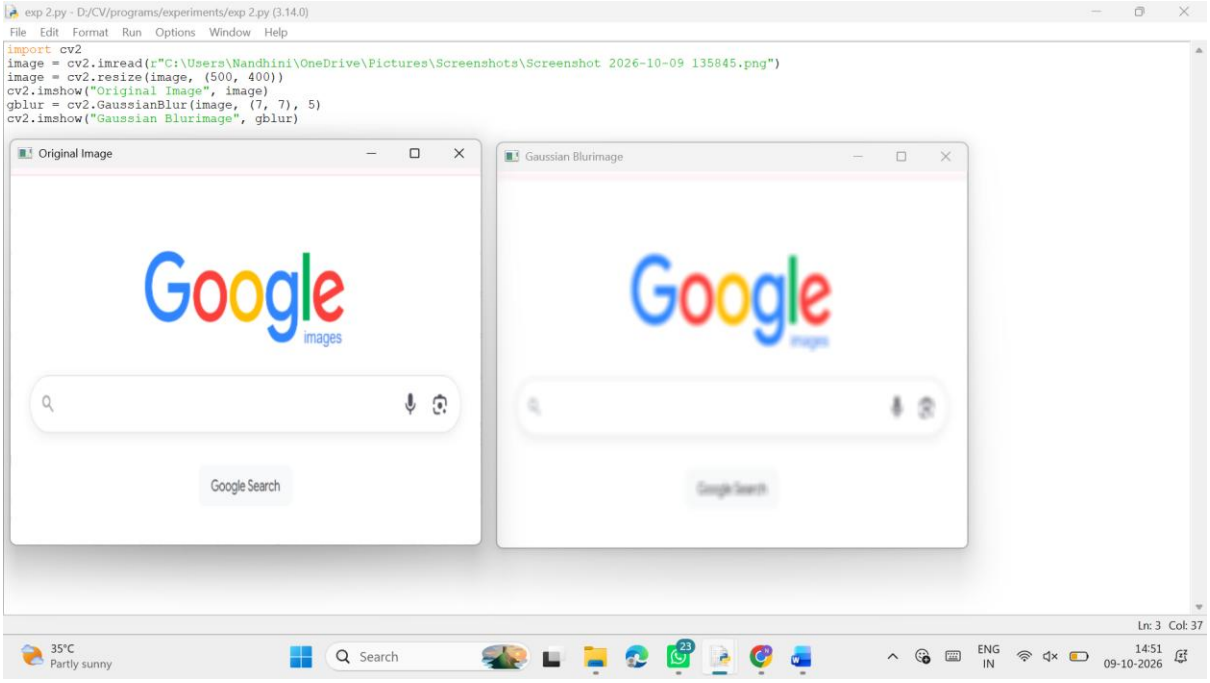
Name:M.Nandhini

Reg.No:192472318

1. Image Reading and Conversion into Gray Scale using OpenCV



2. Image Reading and Conversion into Blur using Gaussian Blur

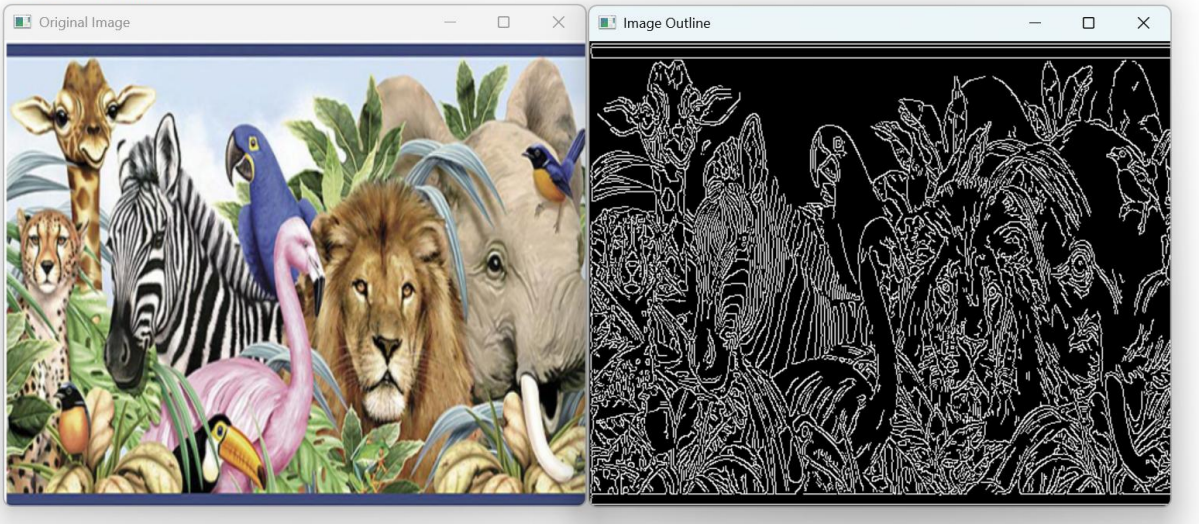


3. Image Outline Detection using Canny Edge Detection

exp 3.py - D:/CV/programs/experiments/exp 3.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray_image, 100, 200)
cv2.imshow("Original Image", image)
cv2.imshow("Image Outline", edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

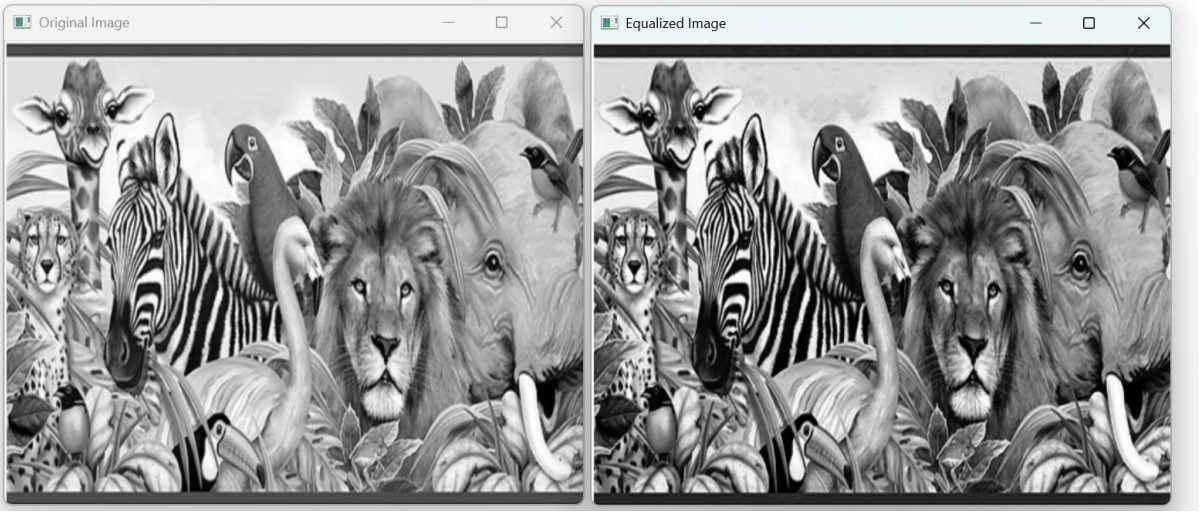


4. Histogram Equalization using OpenCV

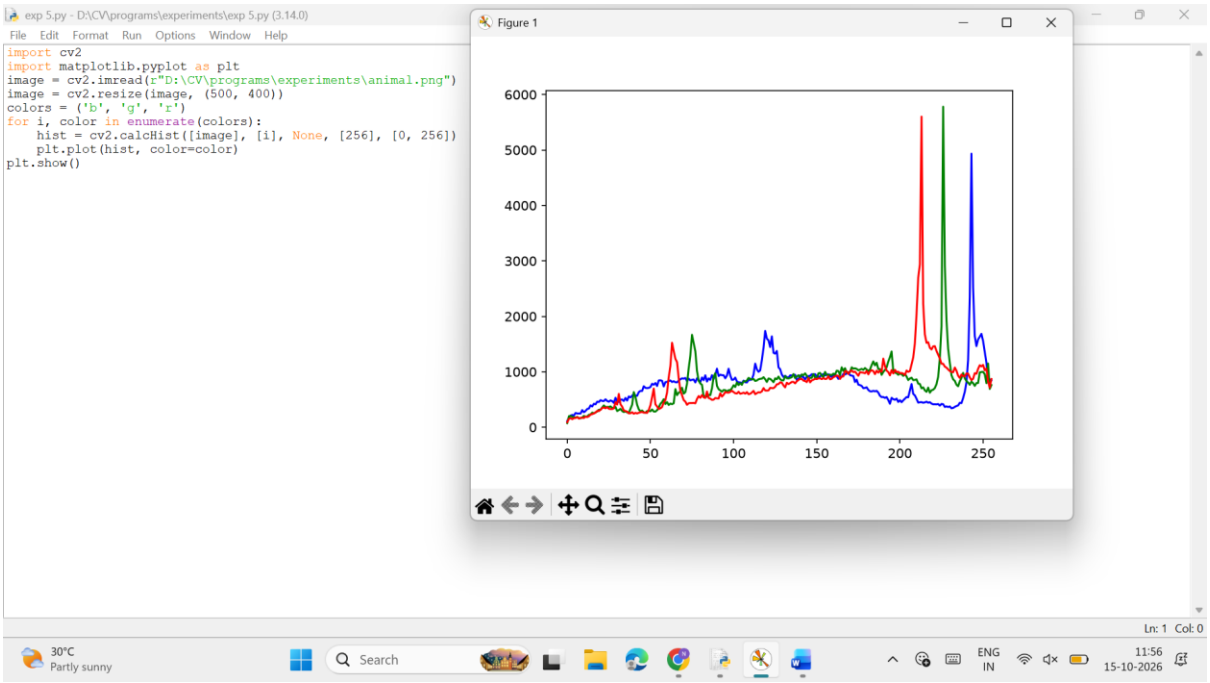
exp 4.py - D:/CV/programs/experiments/exp 4.py (3.14.0)

File Edit Format Run Options Window Help

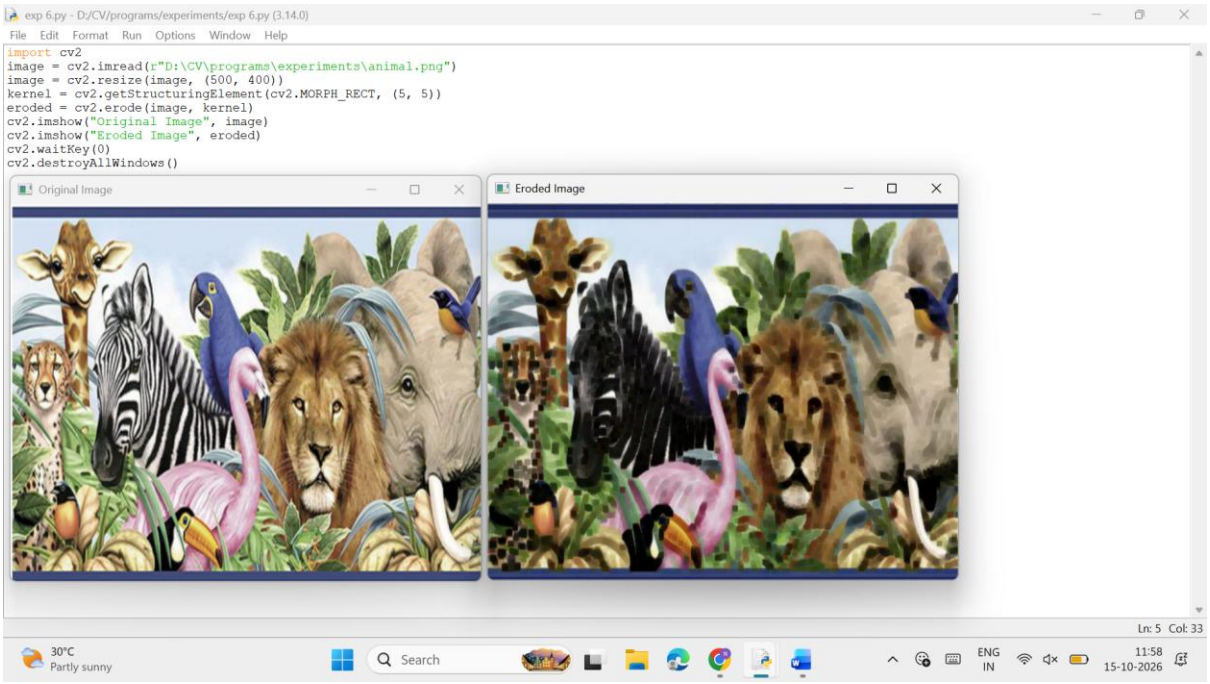
```
import cv2
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
equalized = cv2.equalizeHist(gray_image)
cv2.imshow("Original Image", gray_image)
cv2.imshow("Equalized Image", equalized)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



5. Color Histogram Analysis using OpenCV

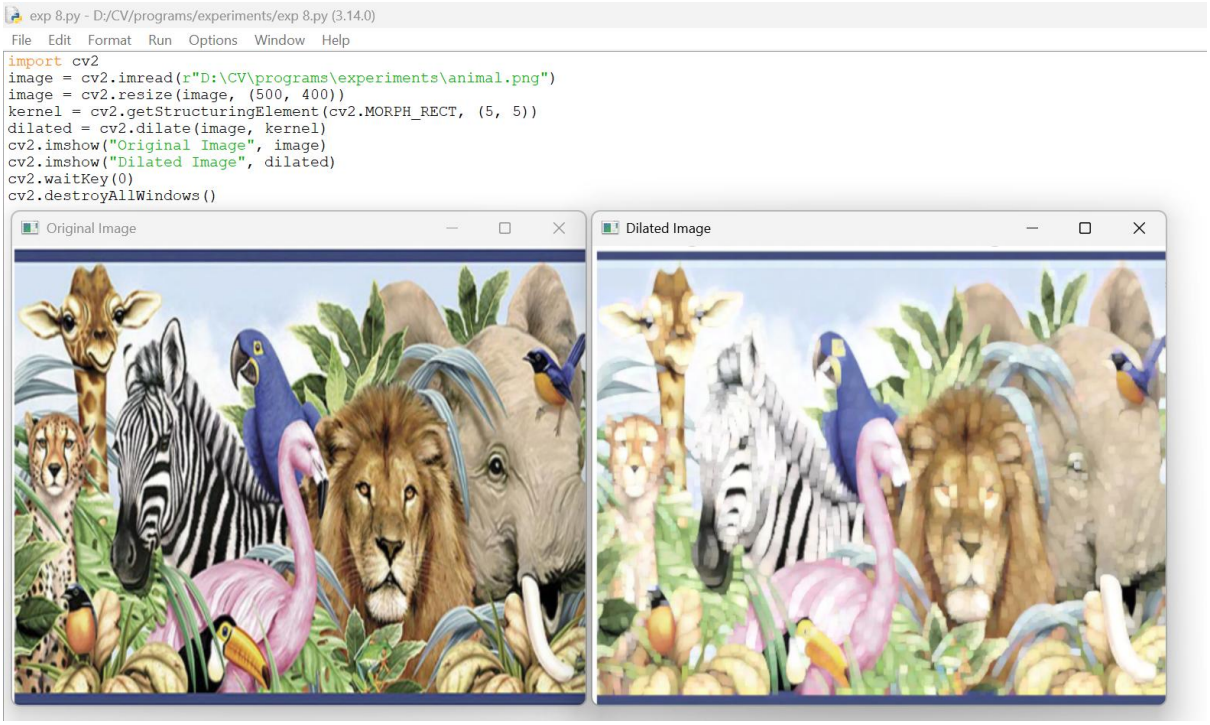


6. Image Erosion using OpenCV

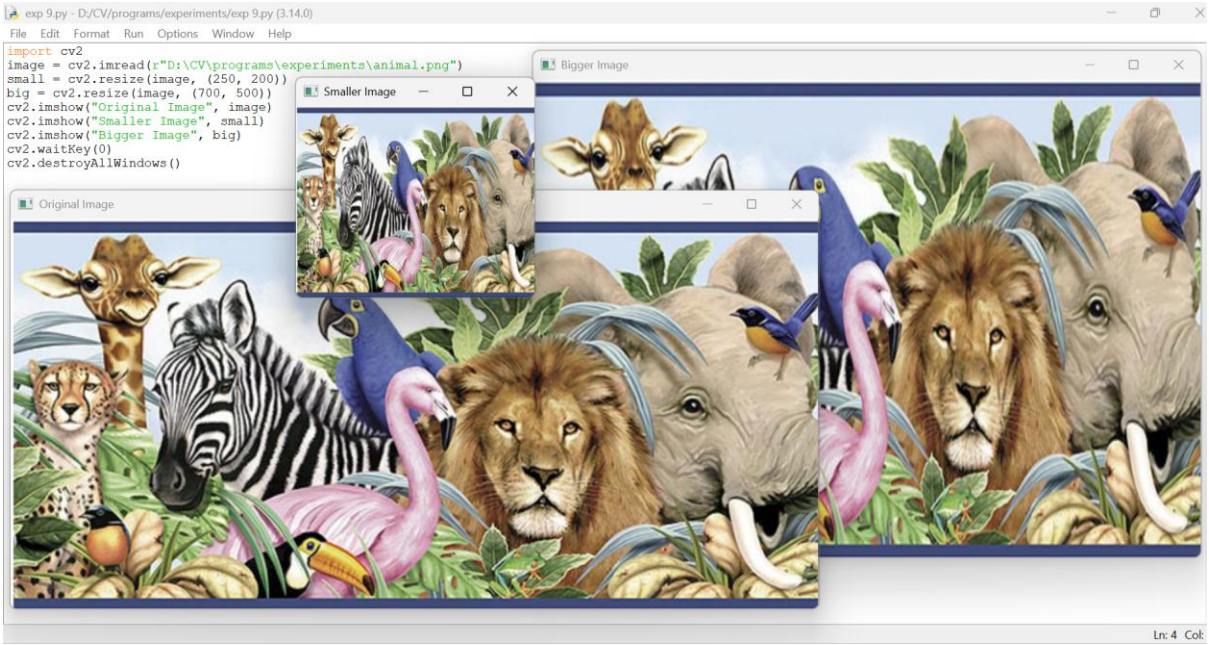


7. Video Processing in Slow Motion and Fast Motion

8. Image Dilation using OpenCV



9. Image Resizing using OpenCV

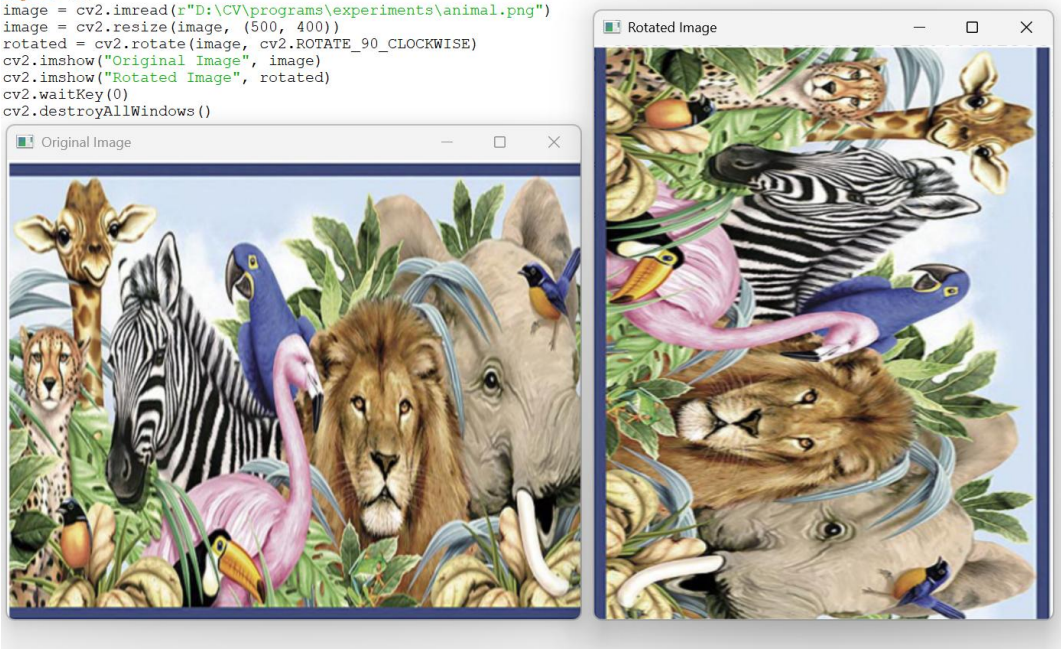


10. Image Rotation by 90 Degrees Clockwise

exp 10.py - D:/CV/programs/experiments/exp 10.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
rotated = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)
cv2.imshow("Original Image", image)
cv2.imshow("Rotated Image", rotated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

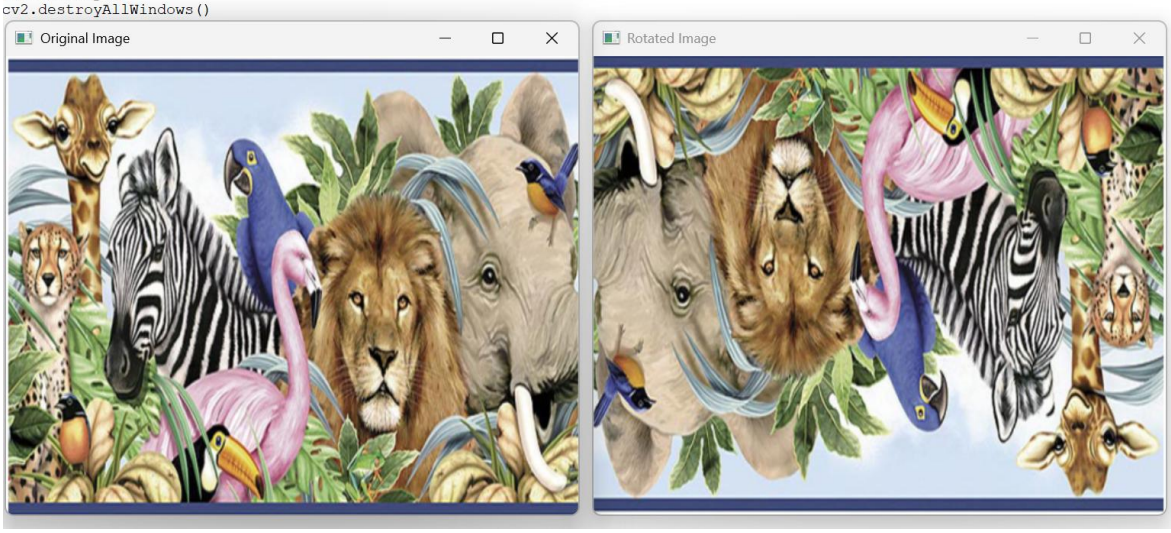


11. Image Rotation by 180 Degrees

exp 11.py - D:/CV/programs/experiments/exp 11.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
rotated = cv2.rotate(image, cv2.ROTATE_180)
cv2.imshow("Original Image", image)
cv2.imshow("Rotated Image", rotated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

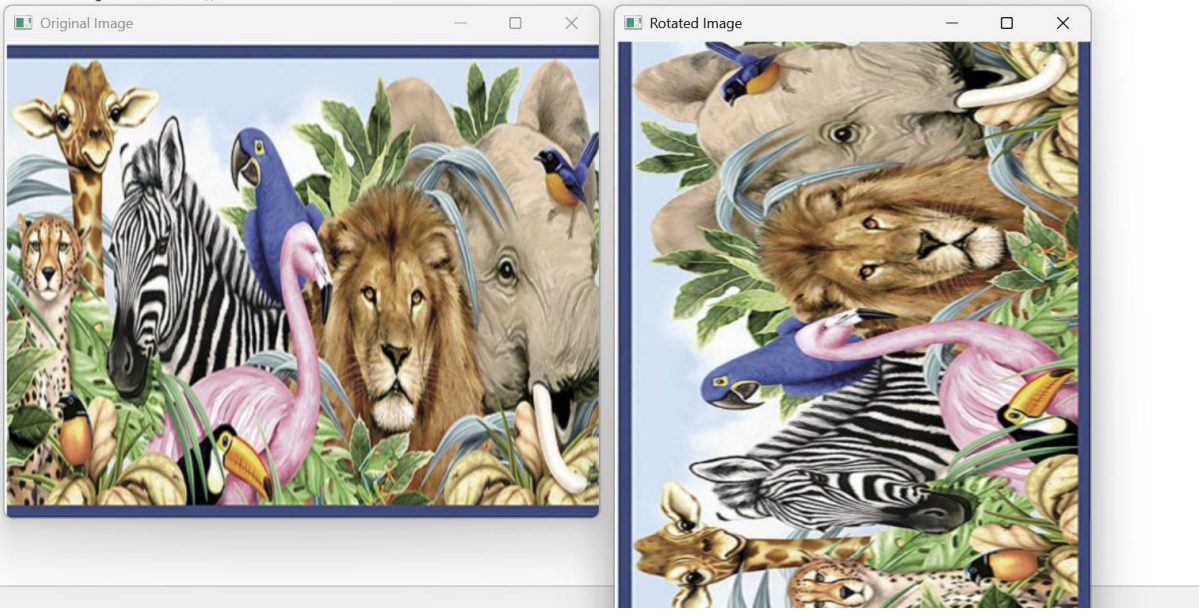


12. Image Rotation by 270 Degrees Clockwise

exp 12.py - D:/CV/programs/experiments/exp 12.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
rotated = cv2.rotate(image, cv2.ROTATE_90_COUNTERCLOCKWISE)
cv2.imshow("Original Image", image)
cv2.imshow("Rotated Image", rotated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

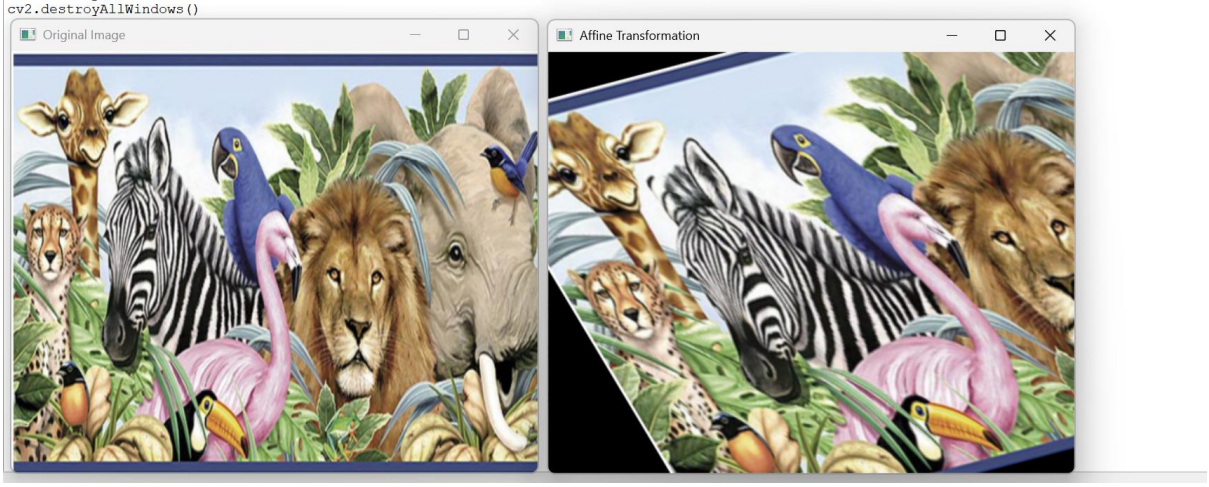


13. Affine Transformation using OpenCV

exp 13.py - D:/CV/programs/experiments/exp 13.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
pts2 = np.float32([[10, 100], [200, 50], [100, 250]])
matrix = cv2.getAffineTransform(pts1, pts2)
result = cv2.warpAffine(image, matrix, (500, 400))
cv2.imshow("Original Image", image)
cv2.imshow("Affine Transformation", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

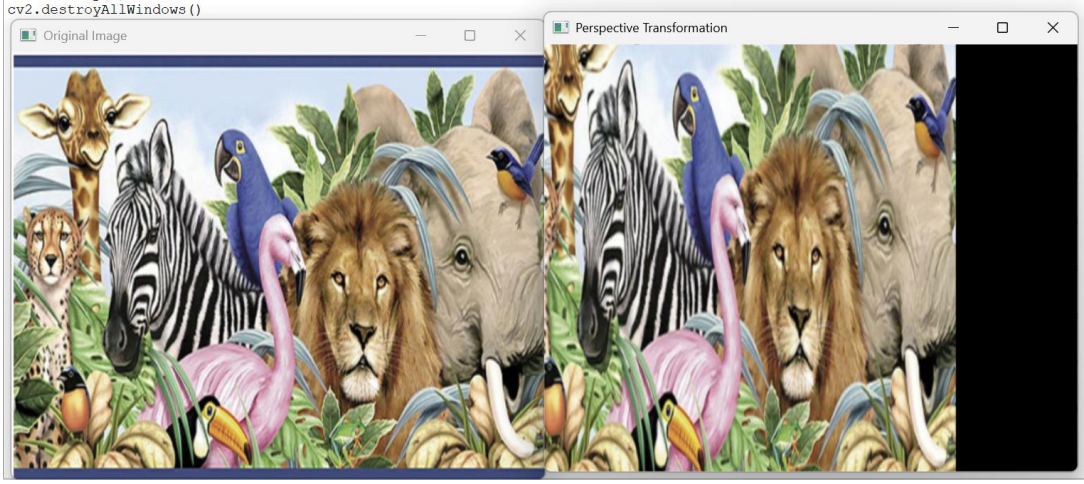


14. Perspective Transformation using OpenCV

exp 14.py - D:/CV/programs/experiments/exp 14.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
pts1 = np.float32([[50, 50], [400, 50], [50, 300], [400, 300]])
pts2 = np.float32([[0, 0], [300, 0], [0, 300], [300, 300]])
matrix = cv2.getPerspectiveTransform(pts1, pts2)
result = cv2.warpPerspective(image, matrix, (500, 400))
cv2.imshow("Original Image", image)
cv2.imshow("Perspective Transformation", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

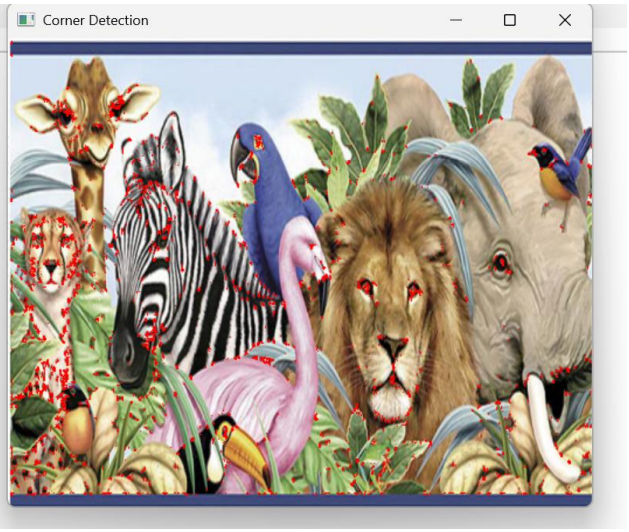


15. Corner Detection using Harris Corner Detection

exp 15.py - D:/CV/programs/experiments/exp 15.py (3.14.0)

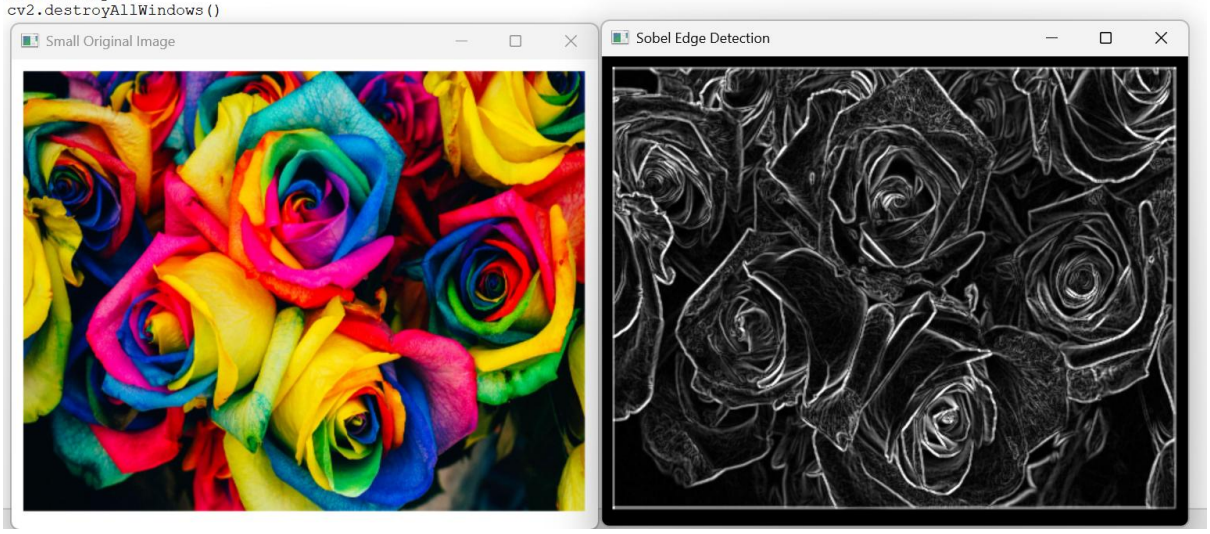
File Edit Format Run Options Window Help

```
import cv2
import numpy as np
image = cv2.imread(r"D:\CV\programs\experiments\animal.png")
image = cv2.resize(image, (500, 400))
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)
corners = cv2.cornerHarris(gray, 2, 3, 0.04)
image[corners > 0.01 * corners.max()] = [0, 0, 255]
cv2.imshow("Corner Detection", image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



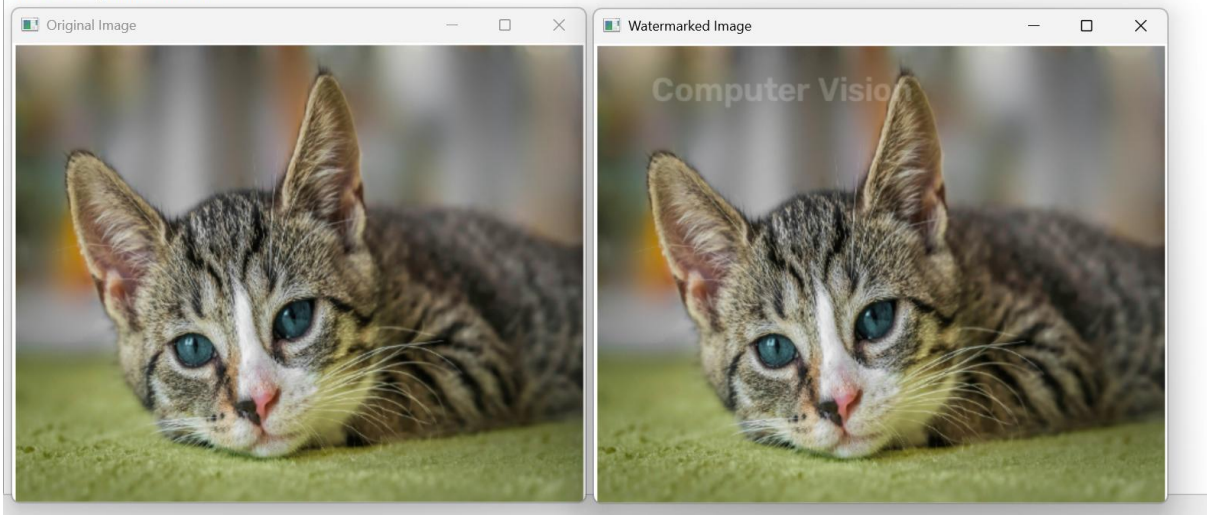
16. Image Filtering using the Sobel Algorithm

```
File  Edit  Format  Run  Options  Window  Help
import cv2
img = cv2.imread(r"D:\CV\programs\experiments\flower.png")
img = cv2.resize(img, (500, 400))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
x = cv2.convertScaleAbs(sobel_x)
y = cv2.convertScaleAbs(sobel_y)
result = cv2.addWeighted(x, 0.5, y, 0.5, 0)
cv2.imshow("Small Original Image", img)
cv2.imshow("Sobel Edge Detection", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

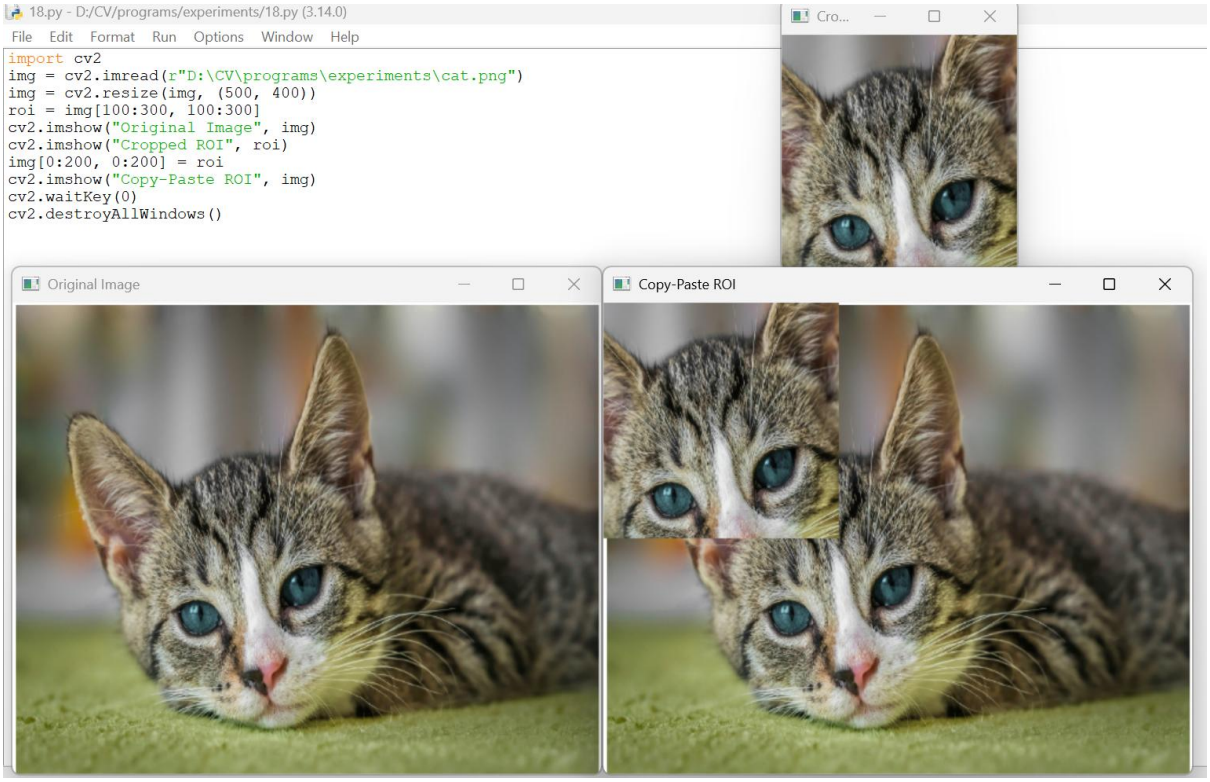


17. Image Watermarking using OpenCV

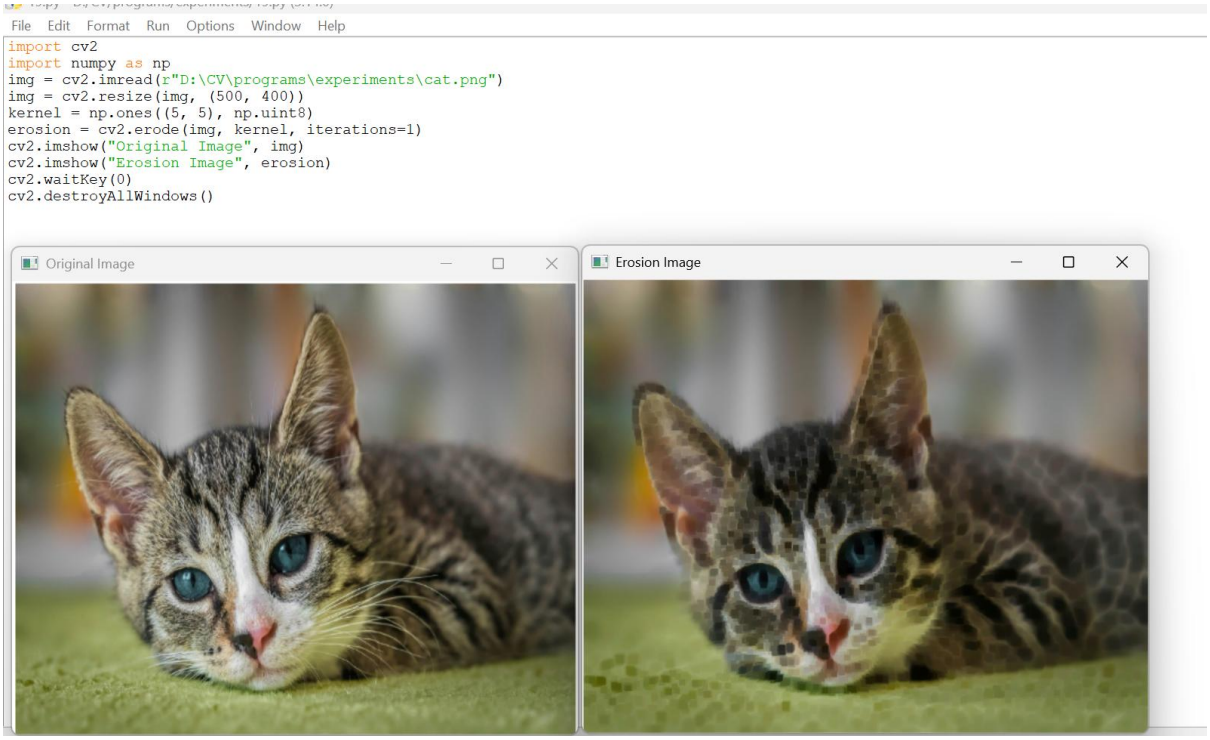
```
exp 17.py - D:/CV/programs/experiments/exp 17.py (3.14.0)
File  Edit  Format  Run  Options  Window  Help
import cv2
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
watermark = img.copy()
cv2.putText(watermark, "Computer Vision", (50, 50),
            cv2.FONT_HERSHEY_SIMPLEX, 1,
            (255, 255, 255), 2)
result = cv2.addWeighted(img, 0.8, watermark, 0.2, 0)
cv2.imshow("Original Image", img)
cv2.imshow("Watermarked Image", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



18. Image Cropping, Copying and Pasting using ROI



19. Morphological Erosion using OpenCV

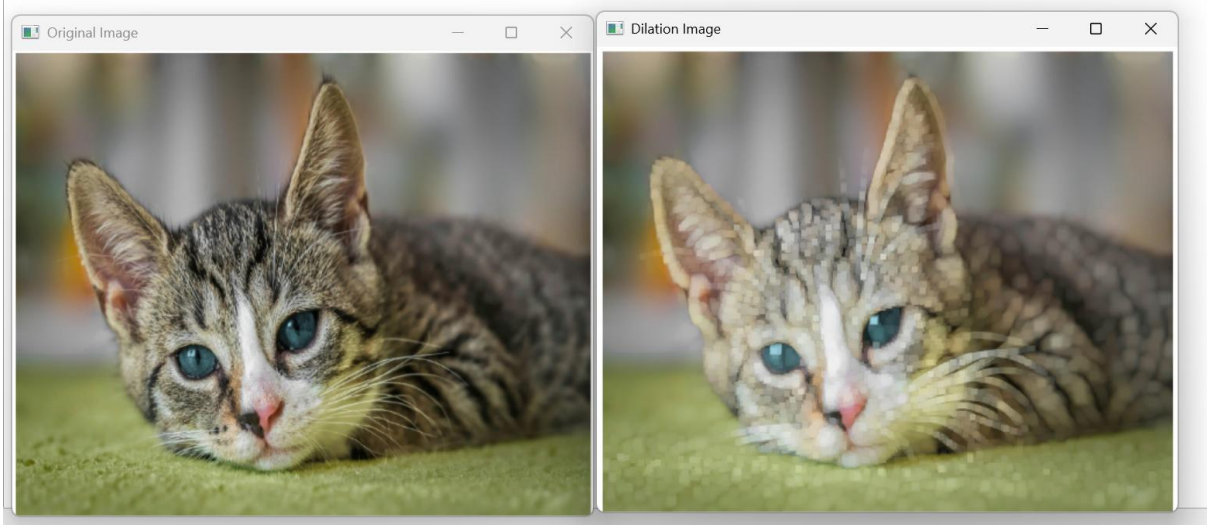


20. Morphological Dilation using OpenCV

20.py - D:/CV/programs/experiments/20.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
kernel = np.ones((5, 5), np.uint8)
dilation = cv2.dilate(img, kernel, iterations=1)
cv2.imshow("Original Image", img)
cv2.imshow("Dilation Image", dilation)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

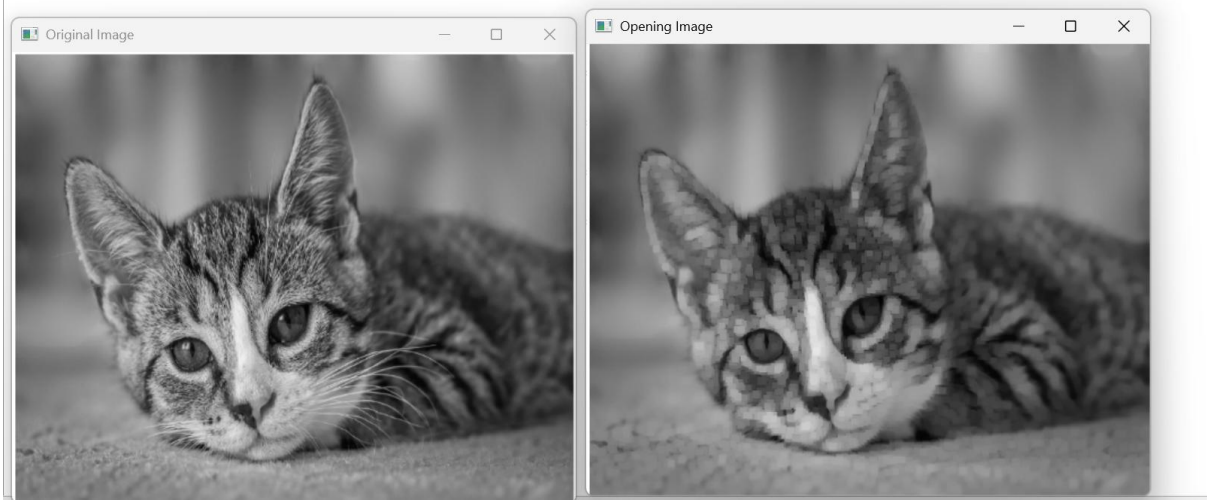


21. Morphological Opening using OpenCV

21.py - D:/CV/programs/experiments/21.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
opening = cv2.morphologyEx(gray, cv2.MORPH_OPEN, kernel)
cv2.imshow("Original Image", gray)
cv2.imshow("Opening Image", opening)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

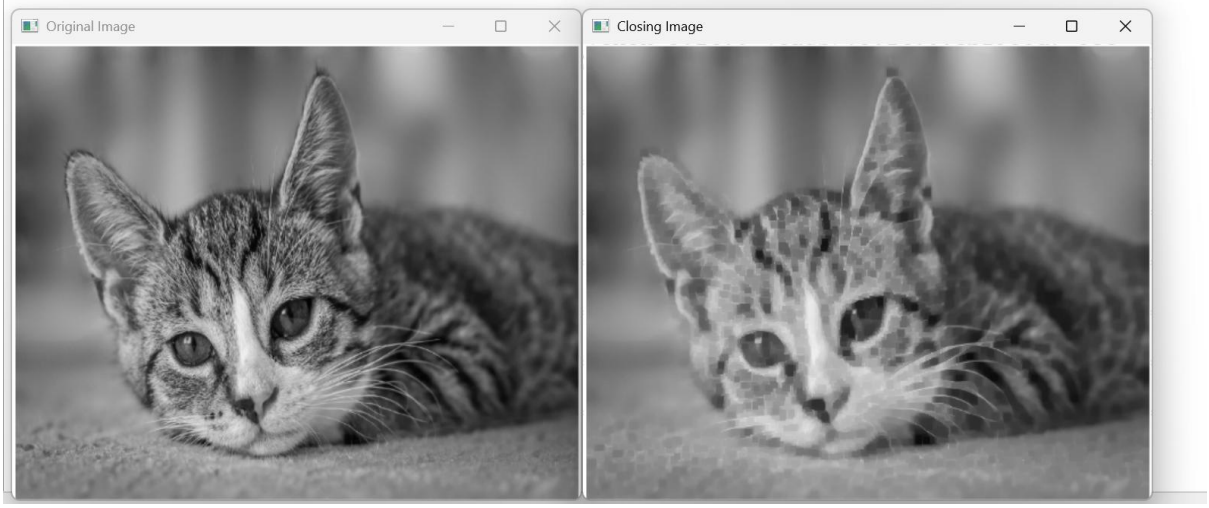


22. Morphological Closing using OpenCV

22.py - D:/CV/programs/experiments/22.py (3.14.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
closing = cv2.morphologyEx(gray, cv2.MORPH_CLOSE, kernel)
cv2.imshow("Original Image", gray)
cv2.imshow("Closing Image", closing)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

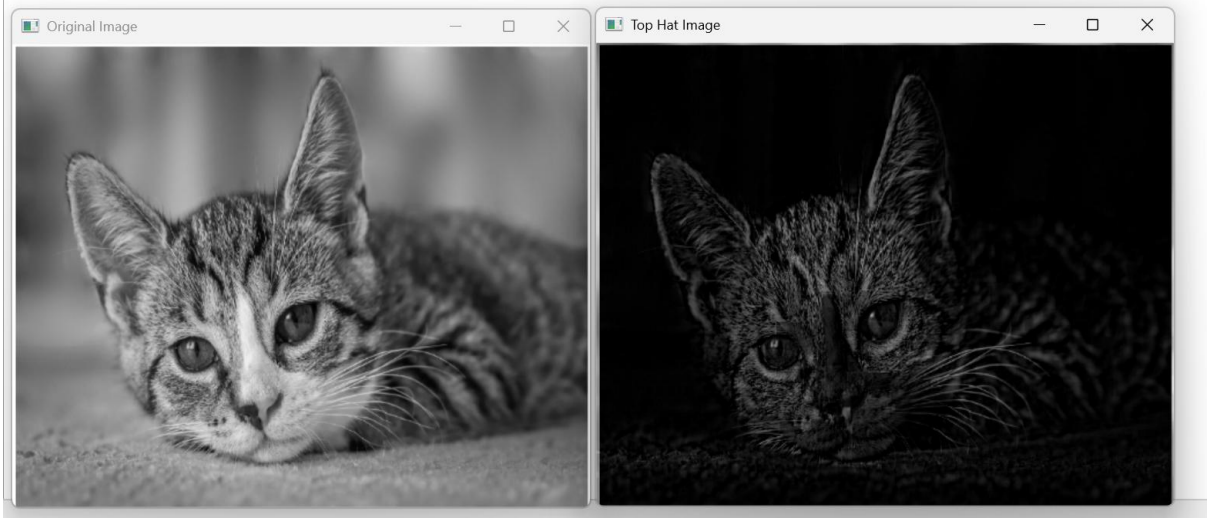


23. Top Hat Morphological Operation using OpenCV

23.py - D:/CV/programs/experiments/23.py (3.14.0)

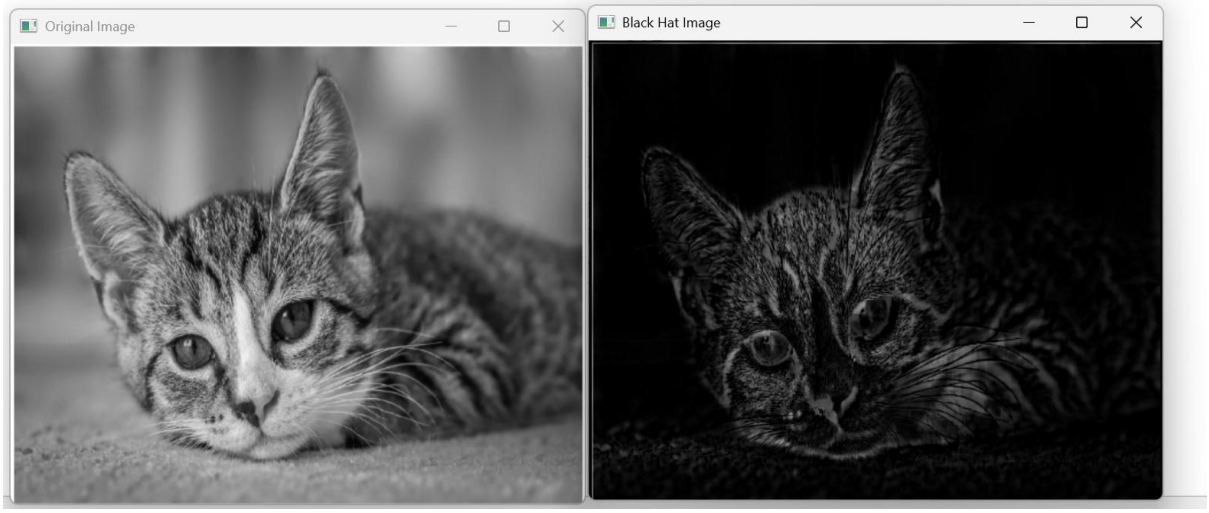
File Edit Format Run Options Window Help

```
import cv2
import numpy as np
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((10, 10), np.uint8)
tophat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)
cv2.imshow("Original Image", gray)
cv2.imshow("Top Hat Image", tophat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



24. Black Hat Morphological Operation using OpenCV

```
24.py - D:/CV/programs/experiments/24.py (3.14.0)
File  Edit  Format  Run  Options  Window  Help
import cv2
import numpy as np
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((10, 10), np.uint8)
blackhat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)
cv2.imshow("Original Image", gray)
cv2.imshow("Black Hat Image", blackhat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

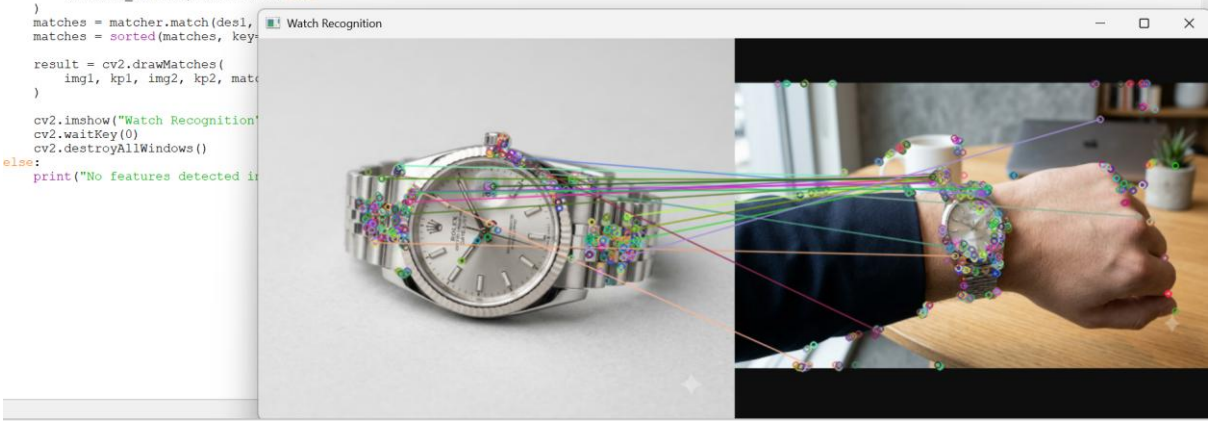


25. Watch Recognition using Object Recognition

```
25.py - D:\CV\programs\experiments\25.py (3.11.0)
File  Edit  Format  Run  Options  Window  Help
import cv2
img1 = cv2.imread(r"D:\CV\programs\experiments\watch.png")
img2 = cv2.imread(r"D:\CV\programs\experiments\test.png")
img1 = cv2.resize(img1, (500, 400))
img2 = cv2.resize(img2, (500, 400))
gray1 = cv2.cvtColor(img1, cv2.COLOR_BGR2GRAY)
gray2 = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)
orb = cv2.ORB_create()
kp1, des1 = orb.detectAndCompute(gray1, None)
kp2, des2 = orb.detectAndCompute(gray2, None)
if des1 is not None and des2 is not None:
    matcher = cv2.BFMatcher(
        cv2.NORM_HAMMING, crossCheck=True
    )
    matches = matcher.match(des1,
    matches = sorted(matches, key

    result = cv2.drawMatches(
        img1, kp1, img2, kp2, mat
    )

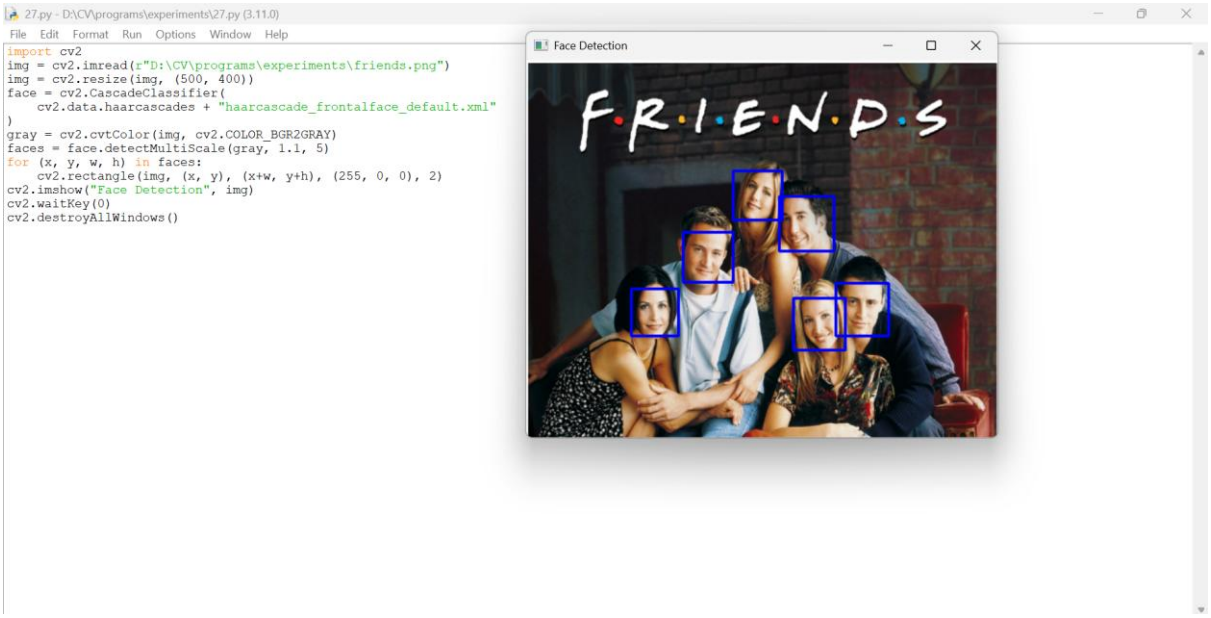
    cv2.imshow("Watch Recognition"
    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("No features detected i
```



26. Reversing Video Frames using OpenCV

26 output.mp4

27. Human Face Detection using OpenCV



28. Vehicle Detection using OpenCV

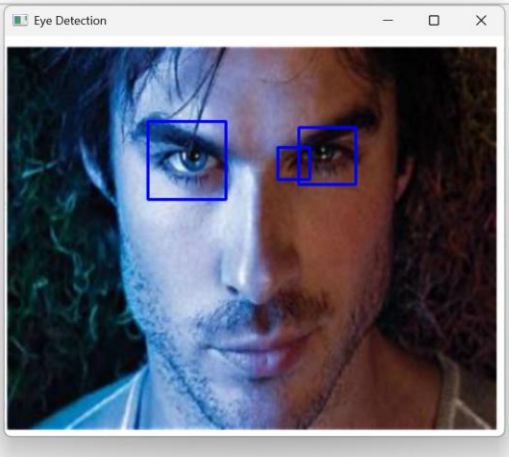
28 output.mp4

29. Eye Detection using OpenCV

29.py - D:/CV/programs/experiments/29.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
eye_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_eye.xml"
)
image = cv2.imread(
    r"D:\CV\programs\experiments\face.jpg"
)
if image is None:
    print("Image not found!")
    exit()
image = cv2.resize(image, (500, 400))
if eye_cascade.empty():
    print("Eye cascade file not loaded!")
    exit()
gray = cv2.cvtColor(
    image,
    cv2.COLOR_BGR2GRAY
)
eyes = eye_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5
)
for (x, y, w, h) in eyes:
    cv2.rectangle(
        image,
        (x, y),
        (x + w, y + h),
        (255, 0, 0),
        2
    )
cv2.imshow("Eye Detection", image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

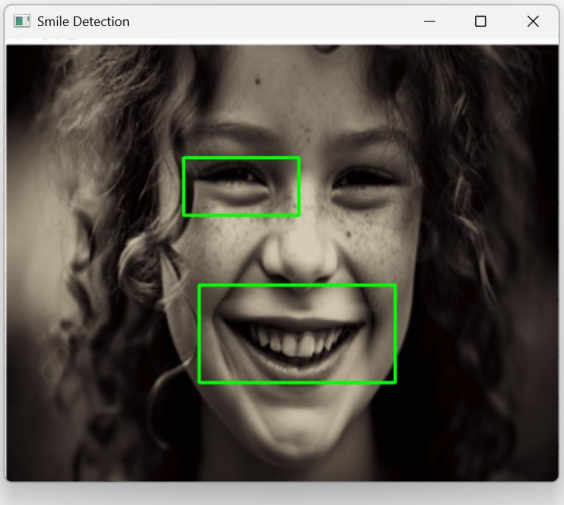


30. Smile Detection using OpenCV

30.py - D:/CV/programs/experiments/30.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)
smile_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_smile.xml"
)
image = cv2.imread(
    r"D:\CV\programs\experiments\smile.jpg"
)
if image is None:
    print("Image not found!")
    exit()
image = cv2.resize(image, (500, 400))
if face_cascade.empty() or smile_cascade.empty():
    print("Cascade file not loaded!")
    exit()
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.3,
    minNeighbors=5
)
for (x, y, w, h) in faces:
    face_region = gray[y:y+h, x:x+w]
    smiles = smile_cascade.detectMultiScale(
        face_region,
        scaleFactor=1.7,
        minNeighbors=20
    )
    for (sx, sy, sw, sh) in smiles:
        cv2.rectangle(
            image,
            (x + sx, y + sy),
            (x + sx + sw, y + sy + sh),
            (0, 255, 0),
            2
        )
cv2.imshow("Smile Detection", image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

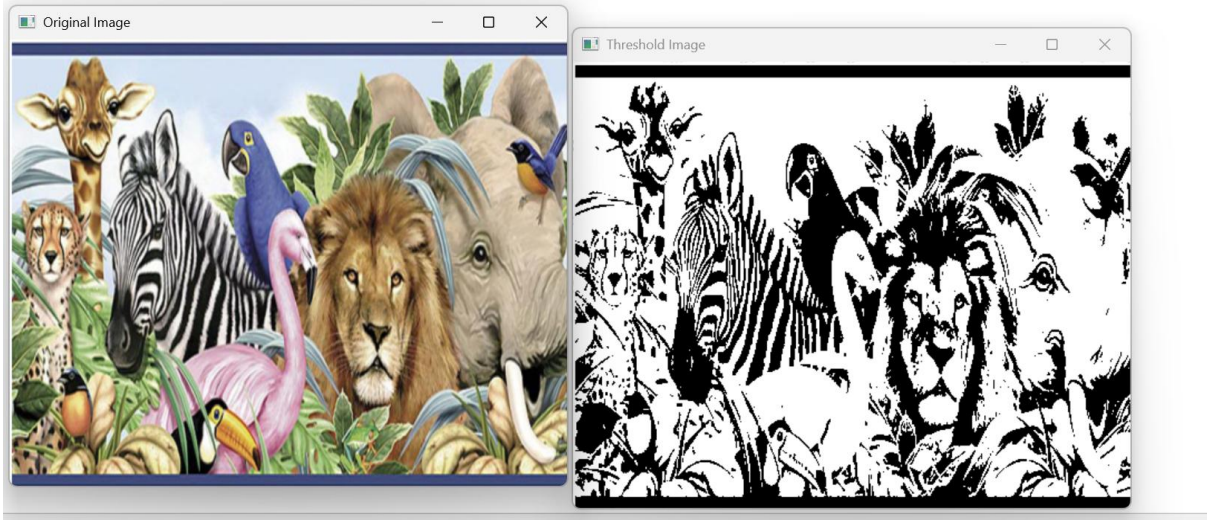


31. Image Segmentation using Thresholding

31.py - D:\CV\programs\experiments\31.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
img = cv2.imread(r"D:\CV\programs\experiments\animal.png")
img = cv2.resize(img, (500, 400))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
ret, thresh = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
cv2.imshow("Original Image", img)
cv2.imshow("Threshold Image", thresh)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

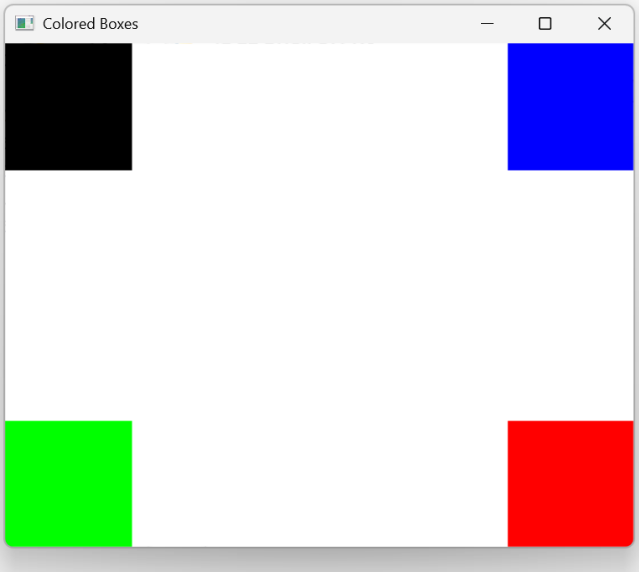


32. Creating a White Image with Four Colored Boxes

32.py - D:/CV/programs/experiments/32.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
img = np.ones((400, 500, 3), dtype=np.uint8) * 255
cv2.rectangle(img, (0, 0), (100, 100), (0, 0, 0), -1)
cv2.rectangle(img, (400, 0), (499, 100), (255, 0, 0), -1)
cv2.rectangle(img, (0, 300), (100, 399), (0, 255, 0), -1)
cv2.rectangle(img, (400, 300), (499, 399), (0, 0, 255), -1)
cv2.imshow("Colored Boxes", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

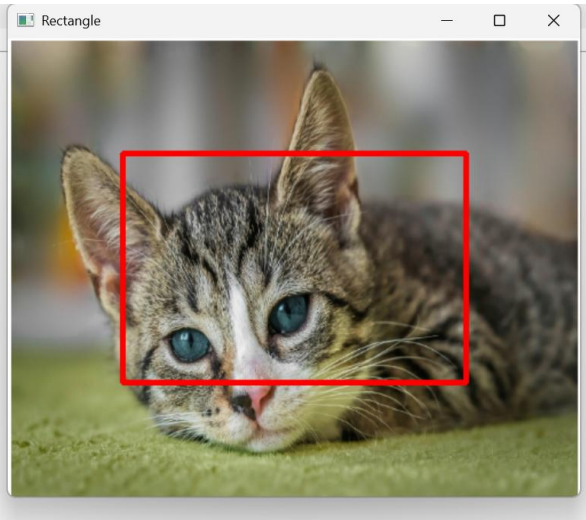


33. Rectangle Drawing using OpenCV

33.py - D:/CV/programs/experiments/33.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
img = cv2.imread(r"D:\CV\programs\experiments\cat.png")
img = cv2.resize(img, (500, 400))
cv2.rectangle(img, (100, 100), (400, 300), (0, 0, 255), 3)
cv2.imshow("Rectangle", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

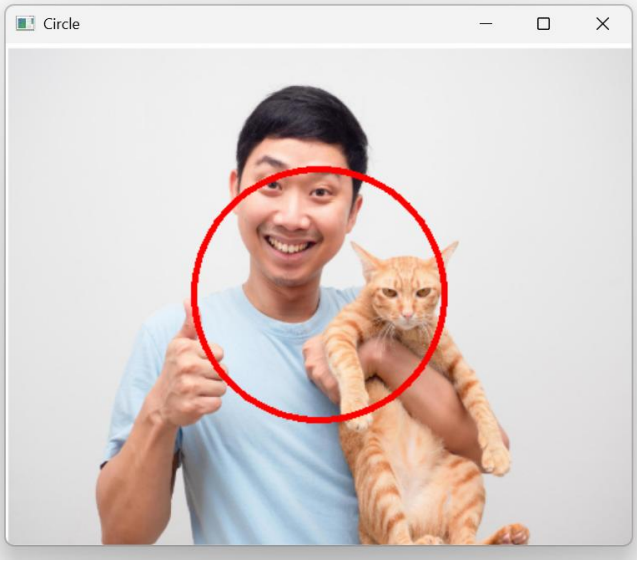


34. Circle Drawing using OpenCV

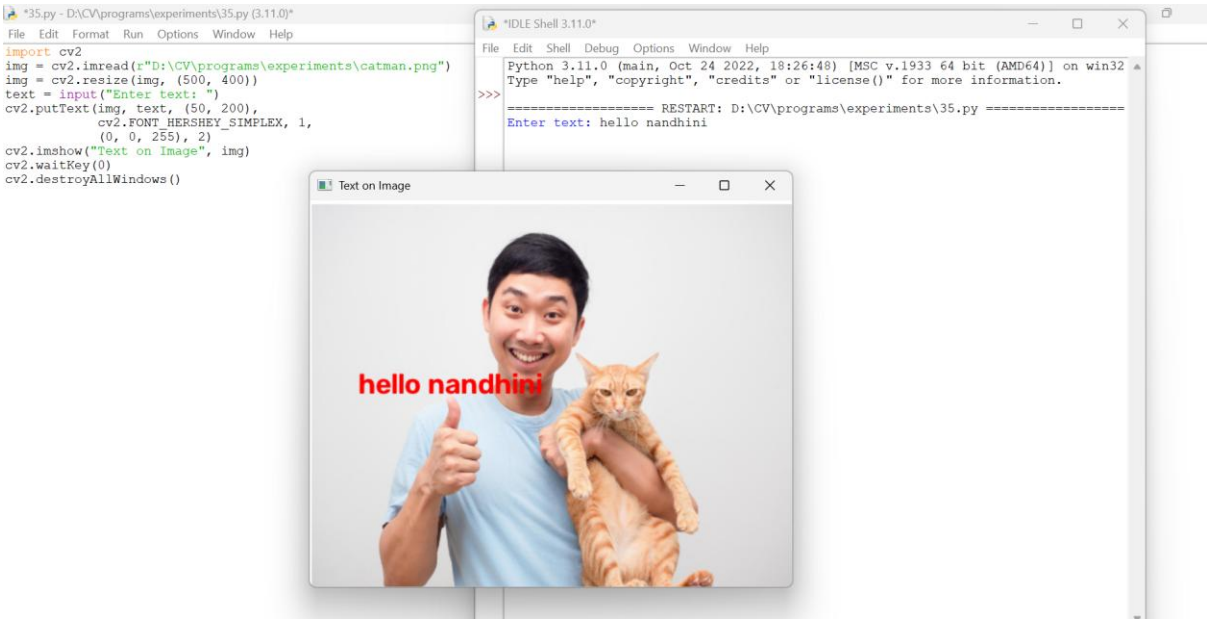
34.py - D:/CV/programs/experiments/34.py (3.11.0)

File Edit Format Run Options Window Help

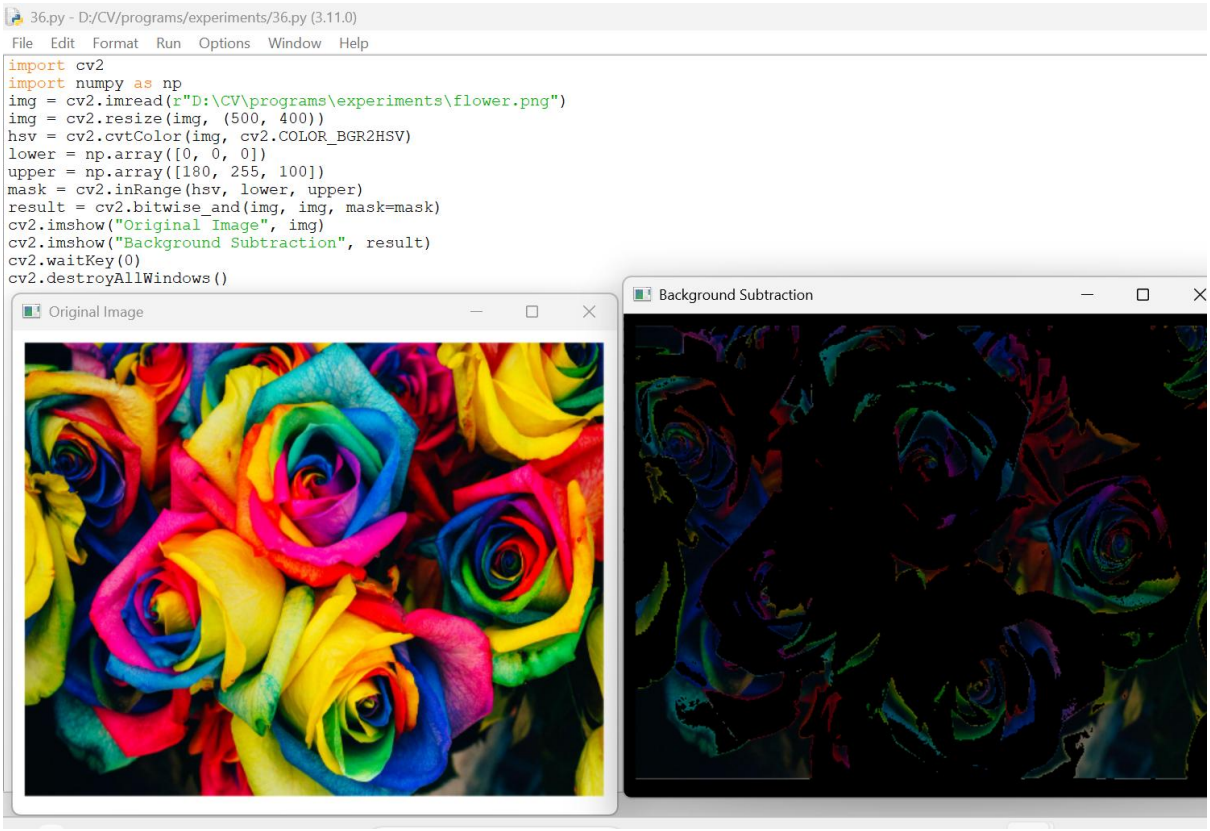
```
import cv2
img = cv2.imread(r"D:\CV\programs\experiments\catman.png")
img = cv2.resize(img, (500, 400))
cv2.circle(img, (250, 200), 100, (0, 0, 255), 3)
cv2.imshow("Circle", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



35. Displaying Text on an Image using OpenCV



36. Background Subtraction based on Color Levels

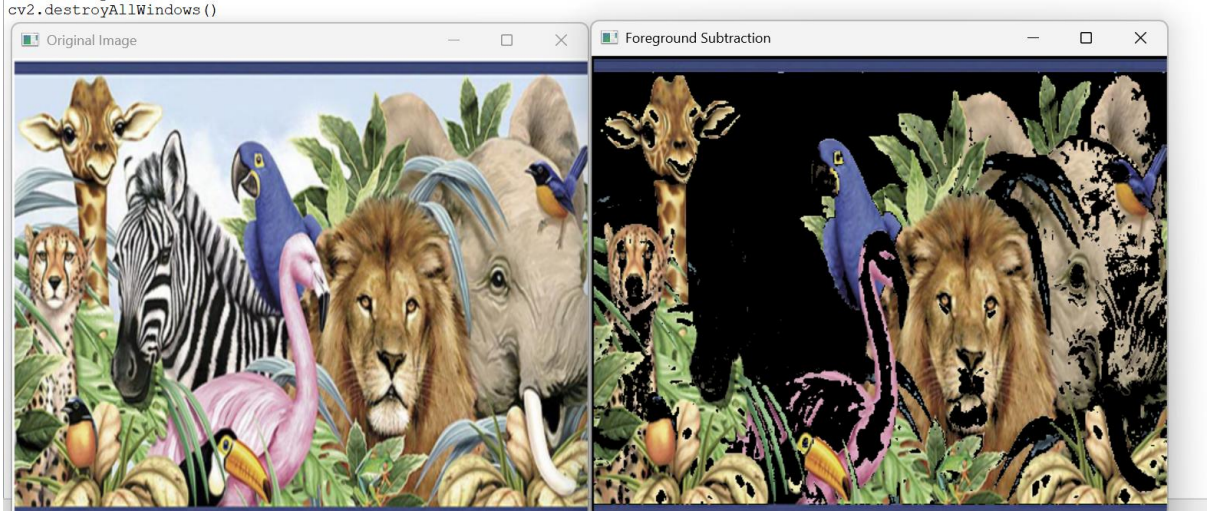


37. Foreground Subtraction based on Color Levels

37.py - D:/CV/programs/experiments/37.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
import numpy as np
img = cv2.imread(r"D:\CV\programs\experiments\animal.png")
img = cv2.resize(img, (500, 400))
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
lower = np.array([0, 50, 50])
upper = np.array([180, 255, 255])
mask = cv2.inRange(hsv, lower, upper)
result = cv2.bitwise_and(img, img, mask=mask)
cv2.imshow("Original Image", img)
cv2.imshow("Foreground Subtraction", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

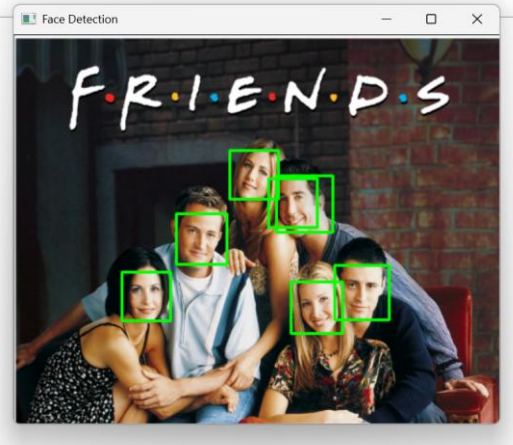


38. Counting Faces in an Image using OpenCV

38.py - D:/CV/programs/experiments/38.py (3.11.0)

File Edit Format Run Options Window Help

```
import cv2
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)
image = cv2.imread(r"D:\CV\programs\experiments\friends.jpg")
if image is None:
    print("Image not found. Check the file name and location.")
elif face_cascade.empty():
    print("Face cascade classifier could not be loaded.")
else:
    image = cv2.resize(image, (500, 400))
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(
        gray,
        scaleFactor=1.1,
        minNeighbors=5
    )
    count = len(faces)
    print("Number of Faces Detected:", count)
    for (x, y, w, h) in faces:
        cv2.rectangle(
            image,
            (x, y),
            (x + w, y + h),
            (0, 255, 0),
            2
        )
    cv2.imshow("Face Detection", image)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```



39. Playing a Video in Reverse Slow Motion



39 output.mp4

40. Extracting Text from Videos using OpenCV



40 output.mp4